Experiment Report

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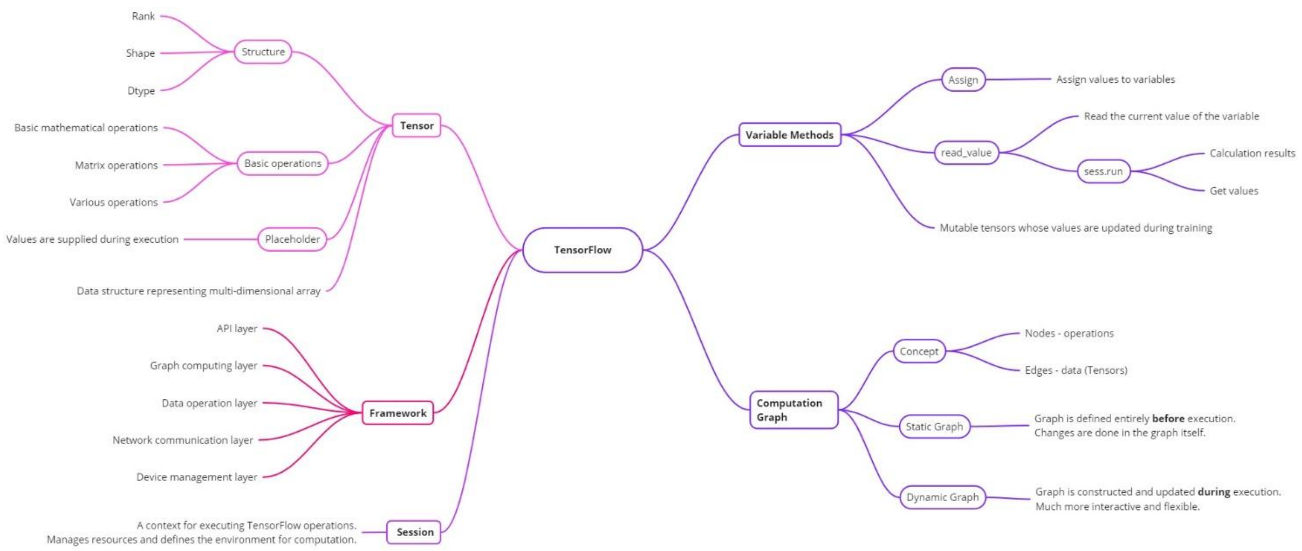
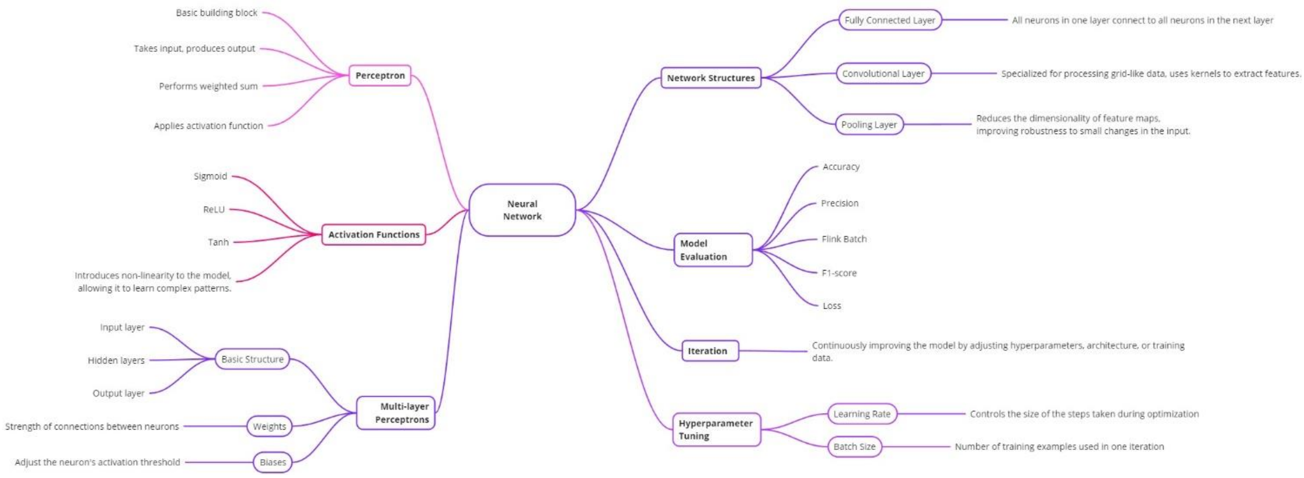
Course: Big Data Analysis Technology

Experiment Topic: TensorFlow

Mind Map TensorFlow



Mind Map Neural Network

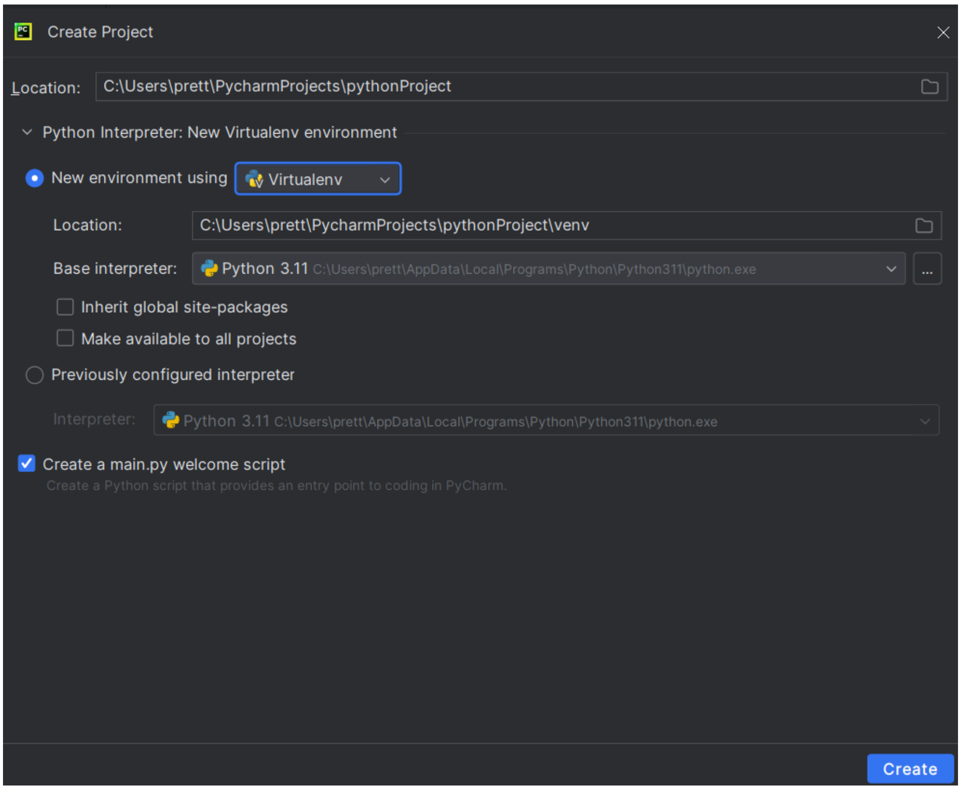


Experiment 0

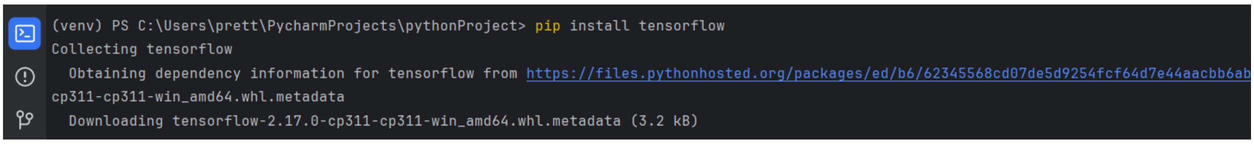
Basic Experiment

Experimental Topic: Training a neural network using the MNIST dataset with Keras.

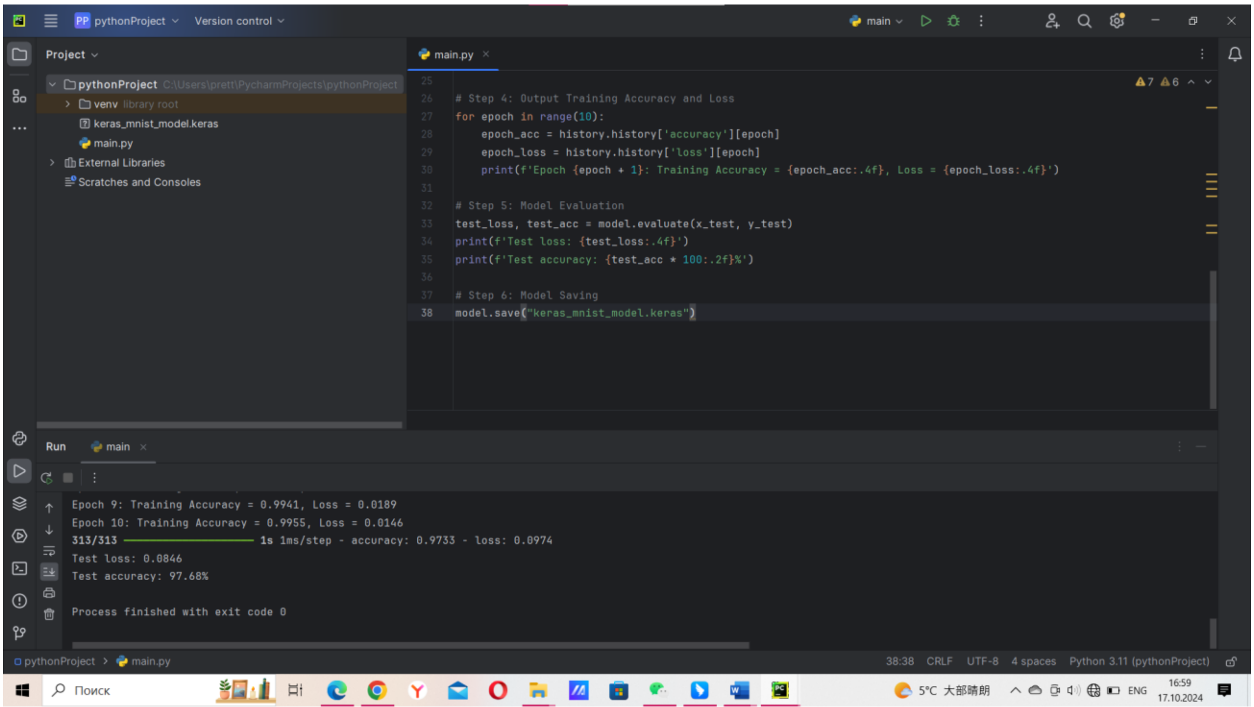
At first, I created a new project with virtual environment using PyCharm.



Then I installed TensorFlow package into my project.



Pasting the code and running it in the program.



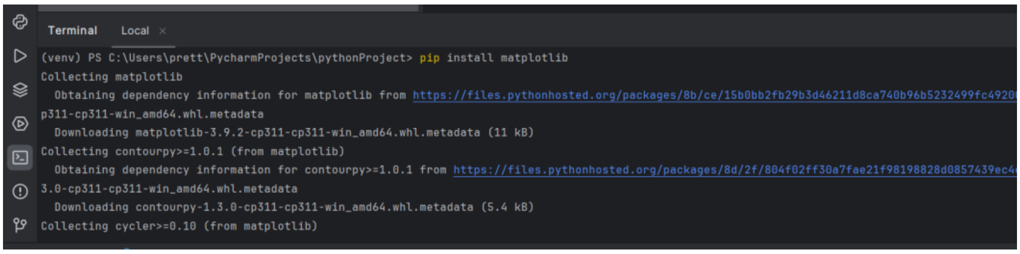
The model showed high performance – with loss 0.0846 and accuracy 97.68%.

Experiment 1

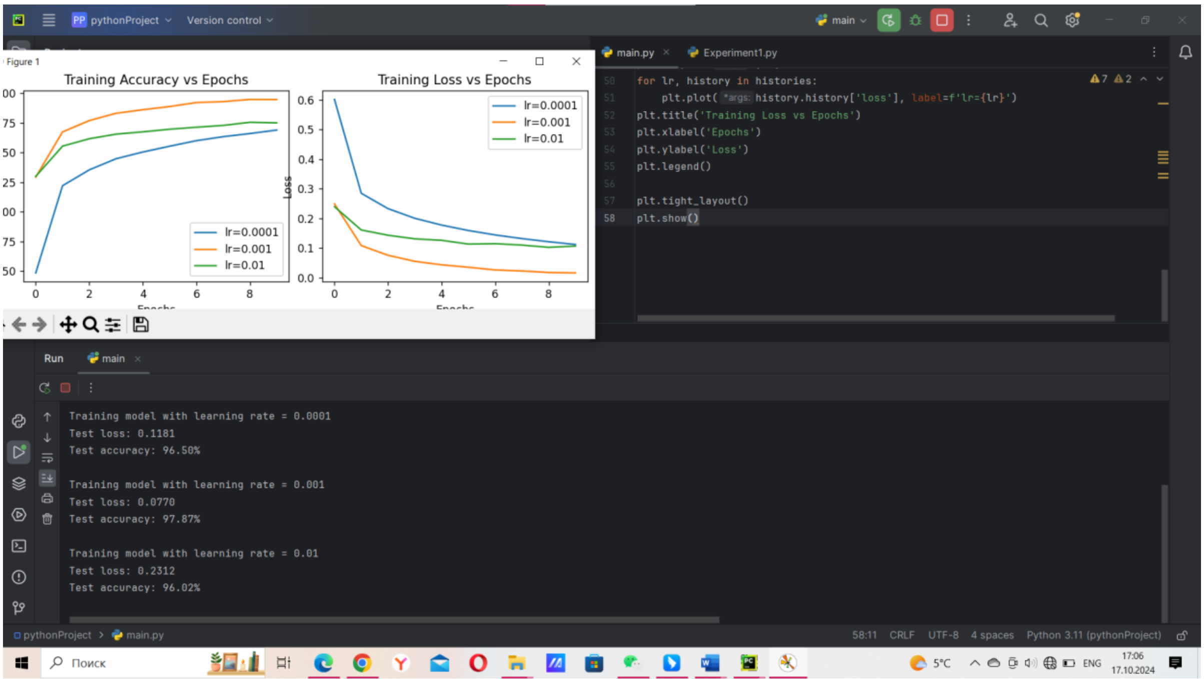
Investigating the Impact of Learning Rate

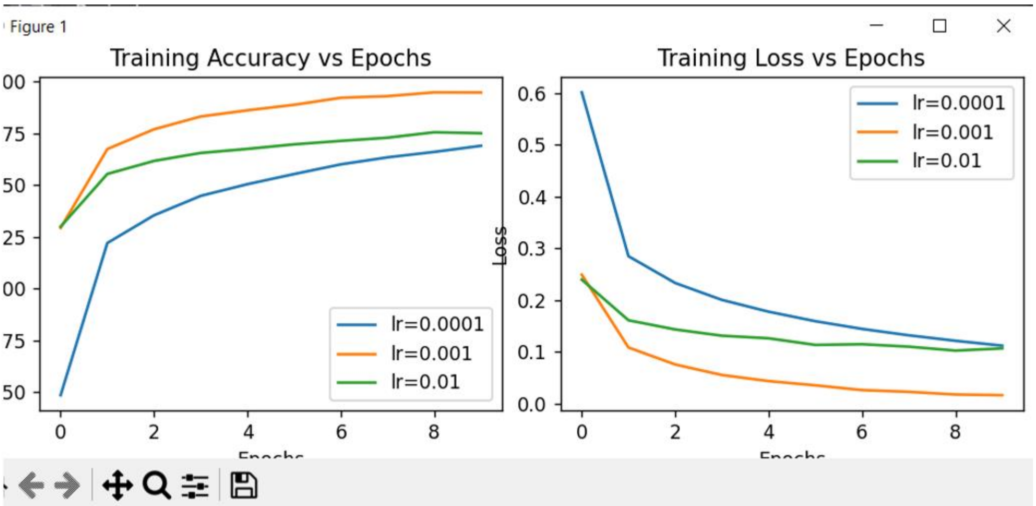
Experimental Topic: Effect of Learning Rate Changes on Neural Network Performance

Installing MatPlotLib



Pasting the code and running it.





Easy to see, that learning rate = 0,001 seems to be the best in this case, as it finds a good balance between speed and accuracy. The blue line is too slow, and the green is too high. So, a smaller learning rate is more stable, but slower. A larger learning rate is faster but can lead to instability.

Experiment 2

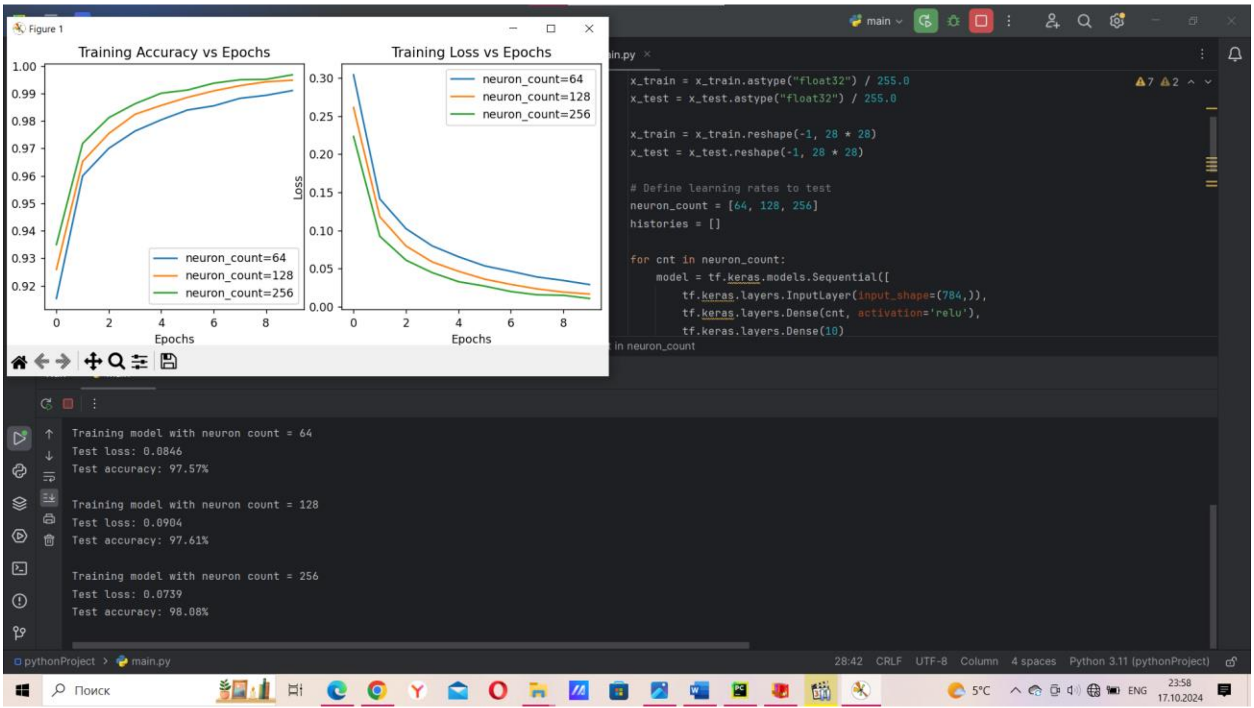
Investigating the Impact of Neuron Count in Hidden Layer

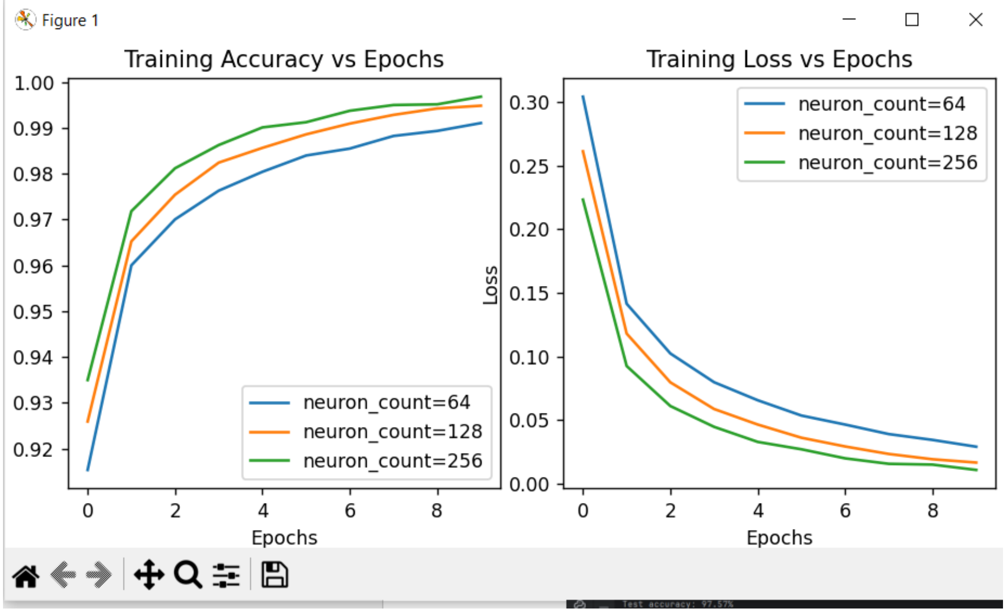
Experimental Topic: Examining the Role of Neuron Counts in Network Architecture

For this experiment it's needed to fix the learning rate at 0.001, modify the number of neurons in the hidden layer with values [64, 128, 256] and record the results (training and test accuracy, and loss). The number of epochs to 10 and a batch size of 32 were the same in the Experiment 1.

For doing that I created a new variable [neuron_count] and changed the dependencies in graphs from learning_rate to neuron_count.

Results showed below.





As we can see from the graph, the green line, with 256 neurons, as it may seem like the best, shows signs of overfitting. It learns the training data too well but might not generalize well to new, unseen data. The orange line, with 128 neurons, seems to strike a good balance between learning speed, accuracy, and avoiding overfitting. This suggests it’s likely the best model in this case.